

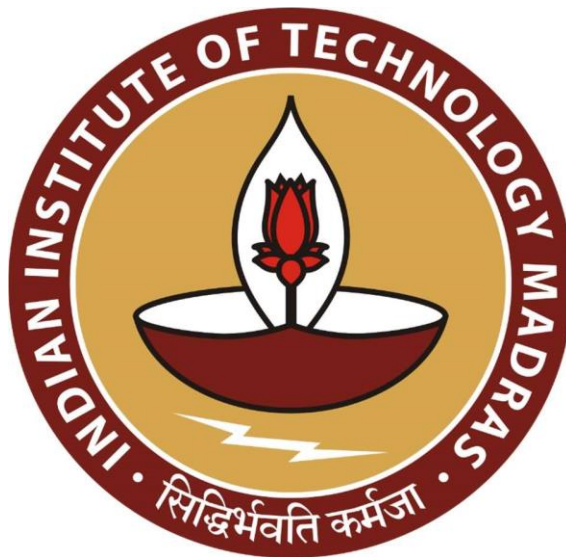
Enhancing Real Estate Business Through Data Driven Analysis

A Proposal report for the BDM capstone Project

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Declaration Statement

I am working on a Project titled “Enhancing Real Estate Business Through Data Driven Analysis”. I extend my appreciation to Narang Associates, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analysed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

signature of Candidate: (Digital Signature)

Name: Manisha Pal

Date: 26 June 2024

2.Executive Summary:

This project focuses on Narang Associates, a small real estate business located in Indira Nagar, Aadrash Nagar, Delhi. The business operates primarily in the B2C sector, dealing with residential property sales and rentals.

Reasons for Choosing This Business:

- Proximity: The business is located in my neighborhood.
- Scale: The size and scope of the business are manageable.
- Cooperation: The owner is willing to share business data.
- Personal Connection: I have known the owner for several years.
- Owner's Education: The owner is educated and understands my proposal.

Key Business Challenges:

- Limited market reach.
- High competition within the current geographic region.
- Lack of data-driven decision-making.

Objectives:

- Extend market reach.
- Optimize marketing strategies.
- Increase profitability.
- Attract new customers.

These objectives will be achieved by analyzing collected data using various analytical methods. Techniques include descriptive statistics to understand current performance, visualization tools to identify trends, and inferential statistics for data-driven decision-making. This project aims to provide strategic insights that will help Narang Associates expand its market reach, optimize marketing efforts, increase overall profitability, and gain new customers.

The anticipated outcomes are enhanced market reach by identifying high-performing regions and potential new markets, improved decision-making through data-driven insights, increased profitability by focusing on high-commission deals, and strategic expansion into new geographical areas. Through this project, Narang Associates will gain a deeper understanding of its business dynamics, leading to more effective growth and sustainability strategies.

3.Organisation Background:

Owner: Mr. Rakesh Narang

Organization: Narang Associates

Address: S-17B, Tent Market, Indira Nagar, Aadrash Nagar, Delhi-110033

Narang Associates is a small real estate firm operational for 10 years, focusing on property sales and rentals. Despite having a strong local presence, the business faces challenges in scaling operations due to limited market reach and the absence of data-driven strategies.

The business model is commission-based, earning 1% on property sales and a 15-day rent commission on rental deals. The company has been operational for a decade, establishing a strong local presence. However, the lack of systematic data collection and analysis, combined with reliance on traditional methods and networks, hinders growth and optimization.

The business operates on a fee-based model, earning commissions from successful property deals. The current challenges include limited market reach, high competition, and the absence of data-driven decision-making. The business relies heavily on the owner's memory and manual record-keeping, making it difficult to identify trends and make strategic decisions. Additionally, the business faces operational and marketing challenges, with no follow-ups with clients and communication gaps between buyers and sellers.

3.Problem Statement:

- Limited marketplace reach leading to a stagnant customer base.
- Inability to make data-driven decisions due to lack of data analysis.
- High competition in the current geographical region reducing profit margins.
- Inability to grow outside the existing geographical area, resulting in less investment and profit.

These issues contribute to nominal income margins and restricted business growth. Addressing these challenges involves analyzing collected data using various analytical techniques to achieve fruitful results. The goal is to provide strategic insights that help extend market reach, optimize marketing strategies, increase overall profitability, and attract new customers.

4.Background of the Problem:

The real estate market in Indira Nagar is highly competitive, making it difficult for Narang Associates to achieve significant growth. Contributing factors include reliance on traditional customer acquisition methods, lack of modern marketing techniques, and absence of data-driven decision-making. Internally, the business lacks the infrastructure for effective data collection and analysis, leading to missed opportunities for optimization and growth. Externally, market saturation hinders the acquisition of new clients and higher profit margins.

Currently, the business relies on the owner's memory and manual record-keeping, making it challenging to identify trends and make strategic decisions. There is also a communication gap between sellers and buyers, leading to missed opportunities and reduced profits. Additionally, the business faces challenges in scaling due to limited market reach and high competition within the current geographic region. Without the use of modern marketing techniques and data-driven decision-making, the business struggles to attract new clients and expand its operations.

The company's reliance on traditional methods and lack of systematic data analysis further exacerbate these challenges. To overcome these issues, a systematic approach to data collection and analysis is essential. This will enable the business to identify trends, make informed decisions, and optimize its operations for better profitability and growth. By leveraging data-driven insights, Narang Associates can strategically expand its market reach, improve customer targeting, and enhance overall business performance.

5.Problem Solving Approach:

To address the challenges faced by Narang Associates, a multi-faceted problem-solving technique will be implemented, leveraging data analysis to drive business growth and optimization.

Methods Used:

- **Descriptive Statistics:** To understand current performance.
- **Data Visualization:** Using bar charts, histograms, and pie charts to visualize deal distributions, commissions, and customer demographics.
- **Inferential Statistics:** Conducting chi-square tests and ANOVA to identify associations and compare means across different deal types.
- **Correlation Analysis:** Exploring relationships between variables like property price and commission earned.
- **Market Analysis:** Identifying high-performing regions and property types to focus marketing efforts.

Data Collection:

Data to be collected includes:

- Deal type (rental or sale)
- Property location
- Property type (flat or plot)
- Purpose (commercial or residential)
- Lead source
- Commission earned
- Expenses incurred
- Number of dealers involved
- Client demographics (age, occupation, income, gender)
- Operational and marketing challenges

Analysis Tools:

- **Excel:** For data processing and visualization.
- **Pivot Tables:** For filtering and computing data.
- **Excel Functions:** Such as VLOOKUP, COUNTIF, and Boolean operators.
- **Data Visualization:** Pie charts, stacked histograms, scatter plots, and heat maps.

Detailed Methods with Justification:

The data collected will be time series data, which allows for the use of various trend lines and scatter plots. By leveraging the power of pivot tables, data can be visualized using bar charts, histograms, and pie charts to illustrate deal distributions, commissions, and customer demographics. Scatter plots will be used to explore relationships among variables, such as

property prices and commissions earned. Financial ratios will be computed to analyze key financial quantities, which can help address the business problems.

Inferential statistics will be employed to conduct chi-square tests to examine associations between categorical variables (e.g., type of deal and location) and ANOVA to compare means of commissions earned across different deal types. Correlation analysis will be used to identify relationships between numerical variables (e.g., property price and commission earned). Market analysis will be conducted to identify high-performing regions and property types, helping to focus marketing efforts. Customer demographics will be analyzed to tailor marketing strategies and improve customer targeting.

Data Analysis:

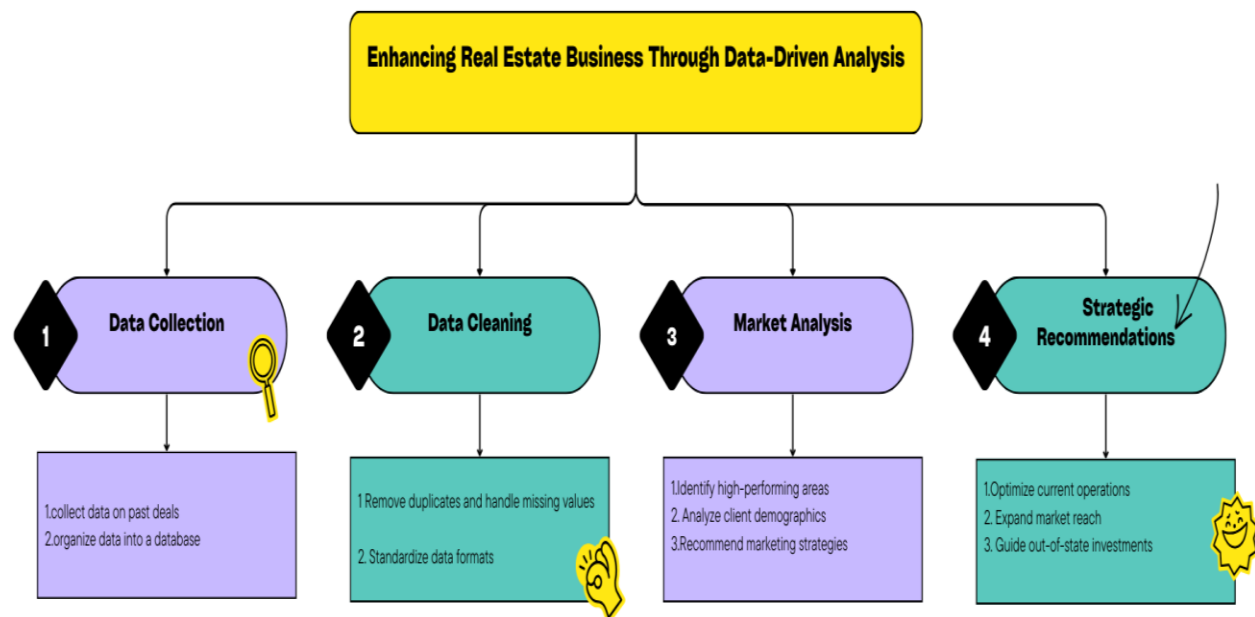
The collected data will be processed and analyzed using Excel. Pivot tables will be used for filtering and computing data, and various Excel functions such as VLOOKUP, COUNTIF, and Boolean operators will be utilized. Data visualization tools such as pie charts, stacked histograms, scatter plots, and heat maps will be employed to provide insights into deal distributions, commissions, and customer demographics.

By leveraging these analytical methods and tools, the project aims to provide strategic insights that will help Narang Associates expand its market reach, optimize marketing strategies, increase overall profitability, and gain new customers.

6.Expected Timeline:

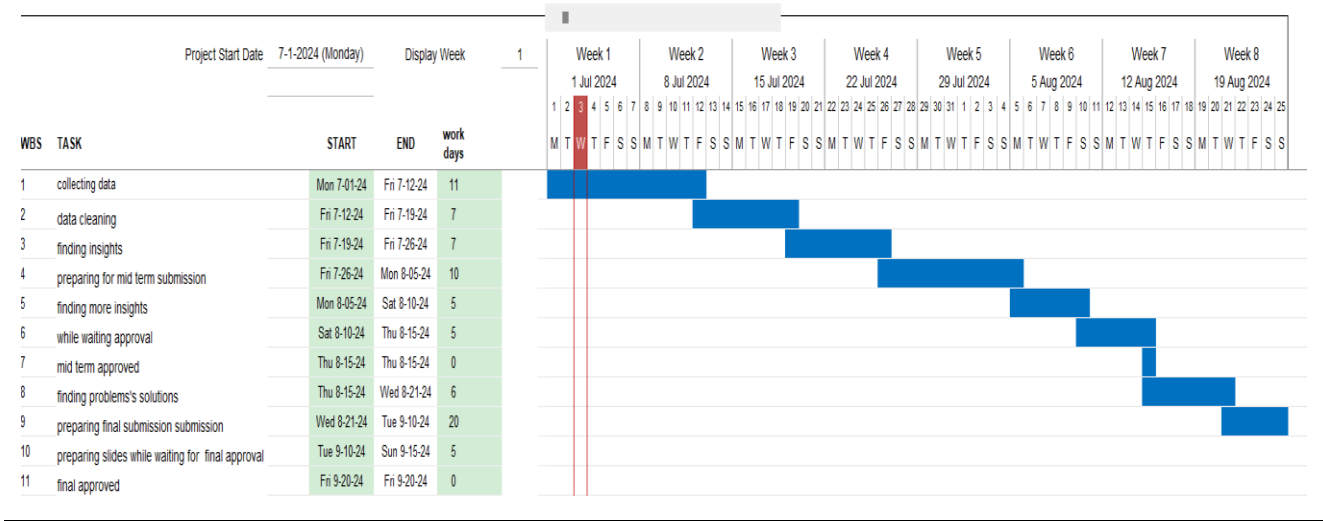
I have successfully collected 3 months of data which includes 4 deals only which will be first cleaned, analysed according to which more data will be collected.

6.1 Work Breakdown Structure:



6.2GANTTCHART:

Enhancing Real Estate Business Through Data Driven Analysis



Expected Outcome:

The anticipated outcomes include:

- **Enhanced Market Reach:** Identifying new markets for expansion.
- **Improved Decision-Making:** Using data-driven insights for strategic decisions.
- **Increased Profitability:** Focusing on high-commission deals and optimizing operations.
- **Strategic Expansion:** Exploring new markets outside the current geographical area.

By leveraging data-driven insights, Narang Associates will gain a deeper understanding of its business dynamics, leading to more effective growth and sustainability strategies. Through this project, the real estate company will be able to extend its market reach, optimize its marketing efforts, increase overall profitability, and attract new customers.